

- NTMP is an integer value corresponding to the number of specified lake boundary conditions to follow. For the first stress period, this value must be greater than or equal to zero, but may be less than zero in subsequent stress periods.

If $NTMP \geq 0$, NTMP is the number of lake boundary conditions.

If $NTMP = -1$, lake boundary conditions from the previous stress period are reused for the current stress period.

(Read item 4 for each NTMP boundary condition)

- 4 Record: ILKBC, ILKBCTYP, (CBCLK(n), n=1, NCOMP)
 Format: Free
- ILKBC is an integer value that is the lake number for which the current boundary condition will be specified
 - ILKBCTYP is an integer value that specifies, for ILKBC, what the boundary condition type is:
 - = 1, a precipitation boundary condition. If precipitation to lakes is simulated in the flow model and a non-zero concentration (default is zero) is associated with it, use $ILKBCTYP = 1$;
 - = 2, a runoff boundary condition. If runoff specified in the LAK package of MODFLOW has a non-zero concentration (default is zero) associated with it, use $ILKBCTYP = 2$. This input is not the same thing as runoff simulated in the UZF1 package and routed to a lake (or stream) using the IRNBND array.
 - CBCLK is the specified concentration for the current boundary condition. One entry (on the same line) per species.

RCT Package

Input to the Chemical Reaction Package is read on unit $INRCT = 8$, which is preset in the main program. The input file is needed only if chemical reactions are simulated. In addition, the option for modeling transport in a dual-domain system is specified through this file.

For each simulation:

- 1 Record: ISOTHM, IREACT, IRCTOP, IGETSC, IREACTION
 Format: 5I10
- ISOTHM is a flag indicating which type of sorption (or dual-domain mass transfer) is simulated:
 - = 0, no sorption is simulated;
 - = 1, Linear isotherm (equilibrium-controlled);
 - = 2, Freundlich isotherm (equilibrium-controlled);
 - = 3, Langmuir isotherm (equilibrium-controlled);
 - = 4, First-order kinetic sorption (nonequilibrium);

=5, Dual-domain mass transfer (without sorption);
=6, Dual-domain mass transfer (with sorption).
=-6, Dual-domain mass transfer (with different sorption coefficients in mobile and immobile domains).

- IREACT is a flag indicating which type of kinetic rate reaction is simulated:
IREACT = 0, no kinetic rate reaction is simulated;
IREACT = 1, first-order irreversible reaction;
IREACT = 2, MONOD kinetic reaction is simulated;
IREACT = 3, first-order chain reaction is simulated.
IREACT = 100, zeroth-order reaction (decay or production)

Note that options 1 and 2 are not intended for modeling chemical reactions between species.

- IRCTOP is an integer flag indicating how reaction variables are entered:

$IRCTOP \geq 2$, all reaction variables are specified as 3-D arrays on a cell-by-cell basis.

$IRCTOP < 2$, all reaction variables are specified as a 1-D array with each value in the array corresponding to a single layer. This option is mainly for retaining compatibility with the previous versions of MT3DMS.

- IGETSC is an integer flag indicating whether the initial concentration for the non-equilibrium sorbed or immobile phase of all species should be read when non-equilibrium sorption (ISOTHM = 4) or dual-domain mass transfer (ISOTHM = 5, 6, or -6) is simulated:

IGETSC = 0, the initial concentration for the sorbed or immobile phase is not read. By default, the sorbed phase is assumed to be in equilibrium with the dissolved phase (ISOTHM = 4), and the immobile domain is assumed to have zero concentration (ISOTHM = 5, 6, or -6).

IGETSC > 0, the initial concentration for the sorbed phase or immobile liquid phase of all species will be read.

- IREACTION is an integer flag to select a reaction module. At least 2 species must be simulated when this option is used. Additional input is needed in needed when this option is used.

If IREACTION=0, no reaction is simulated.

If IREACTION=1, instantaneous EA/ED reaction is simulated between an ED and an EA.
 If IREACTION=2, kinetic reaction is simulated between multiple EAs and EDs.

(Enter 2A if ISOTHM=1, 2, 3, 4, 6, or -6; but not 5; OR if IREACTION=2)

2A Record: RHOB(NCOL,NROW) (one array for each layer)
 Format: RARRAY

- RHOB is the bulk density of the aquifer medium (unit, ML^{-3}).

(Enter 2B if ISOTHM = 5, 6, or -6)

2B Record: PRSITY2(NCOL,NROW) (one array for each layer)
 Format: RARRAY

- PRSITY2 is the porosity of the immobile domain, i.e., the ratio of pore spaces filled with immobile fluids over the bulk volume of the aquifer medium, when the simulation is intended to represent a dual-domain system.

(Enter 2C for each of the NCOMP species if IGETSC > 0)

2C Record: SRCONC(NCOL, NROW) (one array for each layer)
 Format: RARRAY

- SRCONC is the user-specified initial concentration for the sorbed phase of a particular species if ISOTHM = 4 (unit, MM^{-1}). Note that for equilibrium-controlled sorption, the initial concentration for the sorbed phase cannot be specified.
 SRCONC is the user-specified initial concentration for the immobile liquid phase if ISOTHM = 5, 6, or -6 (unit, ML^{-3}).
 SRCONC is not used if ISOTHM = 1, 2, or 3

(Enter 3a for each of the NCOMP species if ISOTHM $\neq$ 0)

3a Record: SP1(NCOL, NROW) (one array for each layer)
 Format: RARRAY

- SP1 is the first sorption parameter. The use of SP1 depends on the type of sorption selected (i.e., the value of ISOTHM):

 For linear sorption (ISOTHM = 1) and nonequilibrium sorption (ISOTHM = 4), SP1 is the distribution coefficient (K_d) (unit, L^3M^{-1}).

 For Freundlich sorption (ISOTHM = 2), SP1 is the Freundlich equilibrium constant (K_f) (the unit depends on the Freundlich exponent a).

 For Langmuir sorption (ISOTHM = 3), SP1 is the Langmuir equilibrium constant (K_l) (unit, L^3M^{-1}).

For dual-domain mass transfer without sorption (ISOTHM = 5), SP1 is not used, but still must be entered.

For dual-domain mass transfer with sorption (ISOTHM = 6), SP1 is also the distribution coefficient (K_d) (unit, L^3M^{-1}).

For dual-domain mass transfer with sorption (ISOTHM = -6), SP1 is the **mobile domain** distribution coefficient (K_{d_m}) (unit, L^3M^{-1}).

(Enter 3b for each of the NCOMP species if ISOTHM = -6)

3b Record: SP1IM(NCOL,NROW) (one array for each layer)
 Format: RARRAY

- SP1IM is the **immobile domain partitioning/distribution coefficient** ($K_{d_{im}}$) (unit, L^3M^{-1}). This option is entered only if ISOTHM = -6, i.e. if a different partitioning coefficient is simulated for the immobile domain.

(Enter 4 for each of the NCOMP species if ISOTHM $\neq$ 0)

4 Record: SP2(NCOL,NROW) (one array for each layer)
 Format: RARRAY

- SP2 is the second sorption or dual-domain model parameter. The use of SP2 depends on the type of sorption or dual-domain model selected:

For linear sorption (ISOTHM = 1), SP2 is read but not used.

For Freundlich sorption (ISOTHM = 2), SP2 is the Freundlich exponent a.

For Langmuir sorption (ISOTHM = 3), SP2 is the total concentration of the sorption sites available (S) (unit, MM^{-1}).

For non-equilibrium sorption (ISOTHM = 4), SP2 is the first-order mass transfer rate between the dissolved and sorbed phases (unit, T^{-1}).

For dual-domain mass transfer (ISOTHM = 5, 6, or -6), SP2 is the first-order mass transfer rate between the two domains (unit, T^{-1}).

(Enter 5 for each species if IREACT > 0)

5 Record: RC1(NCOL, NROW) (one array for each layer)
 Format: RARRAY

- If IREACT = 1 or 3, RC1 is the first-order reaction rate for the dissolved (liquid) phase (unit, T^{-1}). If a dual-domain system is simulated, the reaction rates for the liquid phase in the mobile and immobile domains are assumed to be equal.
- If IREACT = 2 (MONOD kinetics), RC1 is the product of total microbial concentration, M_t (unit, ML^{-3}) and the maximum specific growth rate of the bacterium, u_{max} (unit, T^{-1}).
- If IREACT = 100 (zeroth-order decay or production), RC1 is the zeroth-order reaction rate coefficient for the dissolved (liquid) phase ($ML^{-3}T^{-1}$) (*positive for decay and negative for production*).

If a dual-domain system is simulated, the rate coefficients for the liquid phase in the mobile and immobile domains are assumed equal.

(Enter 6 for each species if IREACT > 0)

- 6 Record: RC2(NCOL, NROW) (one array for each layer)
 Format: RARRAY
- If IREACT = 1 (first-order kinetic reactions) RC2 is the first-order reaction rate for the sorbed phase (unit, T^{-1}). If a dual-domain system is simulated, the reaction rates for the sorbed phase in the mobile and immobile domains are assumed to be equal. Generally, if the reaction is radioactive decay, RC2 should be set equal to RC1, while for biodegradation, RC2 may be different from RC1. Note that RC2 is read but not used, if no sorption is included in the simulation.
 - If IREACT=100 (zeroth-order decay or production), RC2 is the zeroth-order reaction rate coefficient for the sorbed (solid) phase ($MM^{-1}T^{-1}$) (*positive for decay and negative for production*). If a dual-domain system is simulated, the rate coefficients for the liquid phase in the mobile and immobile domains are assumed equal.

(Enter 7 for each species if IREACT = 2)

- 7 Record: RC3(NCOL, NROW) (one array for each layer)
 Format: RARRAY
- RC3 is the half-saturation constant K_s (unit, ML^{-3}). Note that RC3 is read and used only if IREACT = 2 option to simulate MONOD kinetics is invoked.

(Enter 8 if IREACT = 3, one line for each of the NCOMP – 1 species)

- 8 Record: YLD(NCOMP-1)
 Format: F10.0
- YLD is the yield coefficient between species. The first value in the array is for the reaction between species 1 and species 2; the second value for the reaction between species 2 and 3; and so on. Note that YLD is read and used only if IREACT = 3 option to simulate first-order chain reaction is invoked. This option is only available when more than one species are simulated.

(Enter 9a if IREACTION=1)

- 9a Record: IED, IEA, F
 Format: 2I10, F10.0
- IED is the species number representing the electron donor participating in the EA/ED reaction.
 - IEA is the species number representing the electron acceptor participating in the EA/ED reaction.
 - F is the stoichiometric ratio in the simulated equation
 $ED + F*EA \rightarrow \text{Product}$

(Enter 9b if IREACTION=2)

- 9b Record: rec_FileName

Format: A500

- rec_FileName is the name of the input file that provides parameter information relevant to the kinetic reaction module.

rec_FileName

Parameters required for simulating a kinetic reaction are input in a separate file. Below are the input instructions for that file.

0 Record: [#text]
 Format: Free

- This item is optional and can include as many lines as desired, as long as the first character on each line is #. This line is provided for the user to include comments.

1 Record: NED, NEA, NSPECIAL, IFESLD
 Format: Free

- NED is the number of electron donors.
- NEA is the number of electron acceptors.
- NSPECIAL is the number of special cases.
- IFESLD is an integer flag to simulate solid phase Fe^{3+} .

If IFESLD=0, solid phase Fe^{3+} is not simulated.

If IFESLD=1, solid phase Fe^{3+} is simulated.

(Enter 2 NSPECIAL times if NSPECIAL > 0)

2 Record: ISPEC, SPECIAL(ISPEC), EFCMAX
 Format: Free

- ISPEC is the sequential order of the species that is treated as a special case.
- SPECIAL(ISPEC) is the keyword for species number ISPEC. Three possible keywords are as follows:
 - SOLID – The solid phase concentration is used; it is for the iron reduction process. For this case, Fe^{3+} solid phase will be tracked.
 - MAXEC – The method of Lu et al. (1999) to deal with the iron reduction and methanogenesis simulation is used.
 - STORE – This keyword is for the methanogenesis simulation only. If the methane concentration is over the maximum express field capacity (EFC), the additional mass of methane will be stored, and the result will be output as an unformatted file with a name of “MT3D_Ad_methane.UCN”. This option uses the formula developed by Neville and Vlassopoulos (2008).
- EFCMAX is the maximum express field capacity (EFC). If keyword SOLID is used, then this variable is read but not used.

(Enter 3 NEA times)

3 Record: HSC, IC
 Format: Free

- HSC is the half saturation constant.
- IC is the inhibition constants.

(Enter 4 NEA times)

4 Record: DECAYRATE(1:NED)
 Format: Free

- DECAYRATE is the decay rate of each electron acceptor corresponding to each electron donor.

(Enter 5 NEA+NED times)

4 Record: YIELDC(1:NED)
 Format: Free

- YIELDC is the yield coefficient of each component corresponding to each electron donor.

New input requirements

Below is a discussion on input requirements for the revised MT3D-USGS program for the new reaction package. This section describes in general terms, the input variables required to complete a simulation that considers multiple electron donors and electron acceptors, and production of a lower-order ED from the decay of a higher-order ED. The discussion uses a hypothetical system comprising three EDs and five EAs such that $nED = 3$, $nEA = 5$, and $nED + nEA = 8$. In this hypothetical, the three EAs are (1) benzene, (2) MTBE, and (3) TBA. The simulated relationships are as follows: (1) degradation of benzene, without formation of a product; (2) degradation of MTBE with formation of TBA [yield coefficient = 1]; and, (3) degradation of TBA, without formation of a product. This simulation requires that the following inputs be provided:

1. First order decay rates for each ED corresponding to each TEAP
2. Yield coefficients corresponding to:
 - a. The consumption of each EA due to the degradation of each ED
 - b. The production of a lower-order ED from the degradation of a higher-order ED
3. Inhibition constants
4. Half-saturation constants

The inputs for such a simulation are provided as tables, or matrices, in the following order: decay rates, yield coefficients, inhibition constants, and half-saturation constants.

If it is assumed that the half saturation constant expresses the concentration minimum at which any activity can occur for that species – i.e., that a single-valued half-saturation constant applies to each combination of ED and EA - the half-saturation constants can be provided as a vector with dimensions $nED+nEA$. Figure 8 is an example matrix of nED rows and nEA columns that identifies required inputs for the remaining reaction

parameters, and Figure 9 is an example matrix with nED + nEA rows by nED + nEA columns that the user must fill in when simulating multiple EA and ED reactions. Finally, Figure 10 shows a non-square, non-symmetric matrix with nED rows and nED + nEA columns that the user must specify when simulating multiple ED and EA reactions.

Table 2 – A matrix of maximum first order decay rates are required input for simulating multiple EA and ED reactions, an example of which is shown here. Figure 9, below, also shows input requirements for this type of simulation.

	O ₂	NO ₃	FE2	SO ₄	CH ₃
BTEX	Y	Y	Y	Y	Y
MTBE	Y	Y	Y	Y	Y
TBA	Y	Y	Y	Y	Y

Table 3 – A matrix of yield coefficients is required for simulating multiple EA and ED reactions. Although in the general case the matrix could possess nED+nEA rows, on most occasions the matrix will actually possess nED rows. The entry in the corresponding cell indicates whether a value needs to be provided. If a value must be provided, it is the rate of the column species production/consumption due to degradation of 1 unit of the row species. The entries “+”, “-”, and “N” represent production, consumption, and no relationship, respectively.

	BTEX	MTBE	TBA	O ₂	NO ₃	FE2	SO ₄	CH ₃
BTEX	N	N	N	-	-	+	-	+
MTBE	N	N	+	-	-	+	-	+
TBA	N	N	N	-	-	+	-	+
O ₂	N	N	N	N	N	N	N	N
NO ₃	N	N	N	N	N	N	N	N
FE2	N	N	N	N	N	N	N	N
SO ₄	N	N	N	N	N	N	N	N
CH ₃	N	N	N	N	N	N	N	N

Table 4 – A matrix of required inhibition constants that must be specified when simulating multiple EA and ED reactions. Although in the general case the matrix could possess nED rows, on most occasions the matrix will actually possess only one row; that is, each species in the reaction possesses a single inhibition constant.

	BTEX	MTBE	TBA	O ₂	NO ₃	FE2	SO ₄	CH ₃
BTEX	N	N	N	+	+	+	+	+
MTBE	+	N	N	+	+	+	+	+
TBA	+	+	N	+	+	+	+	+